PLC_4 — watch table + how to drive a crane by hand

Send the recipe from the **software** (Send to PLC), then **press the limit switches in this watch table** to tell the crane it has arrived. The software Plant / Crane view updates as you press.

Golden rule

`IOMap` copies `%I → ls_at` every scan. So **press the `%I` ADDRESS** (not the `ls_at` mirror). At every step, press the switch of the place the crane is **heading to**. Watch `sts_state` climb.

---

1) Watch table — CRANE 7A (paste these)

Type the DB tags in the **Name** column; type the `%I` in the **Address** column.

Watch (read only):

```
"Crane7AData".sts_state
"Crane7AData".sts_step
"Crane7AData".cmd_send
"Crane7AData".task_count
"Crane7AData".mv_fwd
"Crane7AData".mv_rev
"Crane7AData".mv_up
"Crane7AData".mv_down
```

Press (modify these `%I`):

```
%I4.7     Parking      (ls_at[0])
%I4.6     Loading      (ls_at[1])
%I4.4     ls_up  (hoist raised)
%I4.5     ls_dip (hoist in tank)
%I6.1     btn_start
%I5.0     1-1A  (station 2)
%I5.1     1-1B  (station 3)
%I5.2     1-2   (station 4)
%I5.3     1-3   (station 5)
%I5.4     1-4A  (station 6)
%I5.5     1-4B  (station 7)
```

1b) Watch table — CRANE 6 (paste these)

Watch (read only):

```
"Crane6Data".sts_state
"Crane6Data".sts_step
"Crane6Data".cmd_send
"Crane6Data".task_count
"Crane6Data".mv_fwd
"Crane6Data".mv_rev
"Crane6Data".mv_up
"Crane6Data".mv_down
```

Press (modify these `%I`):

```
%I0.7     Parking      (ls_at[0])
%I1.6     Loading      (ls_at[1] & ls_at[18], TRVL-4A)
%I0.4     ls_up  (hoist raised)
%I0.5     ls_dip (hoist in tank)
%I2.1     btn_start
%I1.0     1-7B   (station 12)
%I1.1     1-8    (station 13)
%I1.2     1-9    (station 14)
%I1.3     1-10A  (station 15)
%I1.4     1-10B  (station 16)
%I1.5     1-11   (station 17)
```

---

2) Run the recipe `1-3=5 ; 1-4A=3` on crane 7A — press order

Before you send: press (Modify → TRUE) `%I4.7` (Parking) and `%I4.4` (ls_up) — crane starts parked, hoist up. Then in the software hit **Send to PLC**.

`sts_state` shows	Press (Modify → value)	Crane is doing	goes to
10 GO_LOAD	`%I4.6` = TRUE	drove West to Loading	20
20 LOWER_LOAD	`%I4.5` = TRUE	hoist down at Loading	30
30 WAIT_BTN	`%I6.1` = TRUE (hold ~3 s) then FALSE	operator loads part	40
40 LIFT_LOAD	`%I4.5` = FALSE, `%I4.4` = TRUE	hoist up	50
50 GO_TANK (task 1)	`%I4.6` = FALSE, `%I5.3` = TRUE	drove to **1-3**	60
60 LOWER_TANK	`%I4.4` = FALSE, `%I4.5` = TRUE	dips into 1-3	70
70 DWELL (5 s)	*(wait 5 s)*	soaking in 1-3	80
80 LIFT_TANK	`%I4.5` = FALSE, `%I4.4` = TRUE	lifts out, task 1 done	50
50 GO_TANK (task 2)	`%I5.3` = FALSE, `%I5.4` = TRUE	drove to **1-4A**	60
60 LOWER_TANK	`%I4.4` = FALSE, `%I4.5` = TRUE	dips into 1-4A	70

`sts_state` shows	Press (Modify → value)	Crane is doing	goes to
70 DWELL (3 s)	*(wait 3 s)*	soaking in 1-4A	80
80 LIFT_TANK	`%I4.5` = FALSE, `%I4.4` = TRUE	lifts out, all tasks done	85
85 GO_PARK	`%I5.4` = FALSE, `%I4.7` = TRUE	drove East back to Parking	90
90 DONE	*(software auto-acks)*	finished	0

Pattern: **turn OFF the switch you just left, turn ON the switch you're arriving at.** Only one tank switch TRUE at a time. As you flip them, the crane marker moves on the software Plant/Crane view.

> Tip: if a `%I` modify won't stick while running, use the **Force table** (same addresses) — PLCSIM holds forced inputs solidly.

---

3) Full switch reference (both cranes)

Station = tank number + 1 · `ls_at[]` index = station · press the `%I`.

Crane 7A — `Crane7AData` (entry crane; stations 0,1,2–7)

Tank / where	Station	`ls_at[]`	`%I`
Parking	0	`ls_at[0]`	`%I4.7`
Loading (LOAD-4A)	1	`ls_at[1]`	`%I4.6`
1-1A	2	`ls_at[2]`	`%I5.0`
1-1B	3	`ls_at[3]`	`%I5.1`
1-2	4	`ls_at[4]`	`%I5.2`
1-3	5	`ls_at[5]`	`%I5.3`
1-4A	6	`ls_at[6]`	`%I5.4`
1-4B	7	`ls_at[7]`	`%I5.5`
hoist up	—	`ls_up`	`%I4.4`
hoist in tank	—	`ls_dip`	`%I4.5`
start button	—	`btn_start`	`%I6.1`

Outputs (read): `mv_up`=`%Q2.4` `mv_down`=`%Q2.5` `mv_rev`(East)=`%Q2.6` `mv_fwd`(West)=`%Q2.7`.

Crane 6 — `Crane6Data` (downstream crane; stations 0,1,12–18)

Tank / where	Station	`ls_at[]`	`%I`
Parking	0	`ls_at[0]`	`%I0.7`
Loading (TRVL-4A)	1	`ls_at[1]`	`%I1.6`

Tank / where	Station	ls_at[]	%I
1-7B	12	ls_at[12]	%I1.0
1-8	13	ls_at[13]	%I1.1
1-9	14	ls_at[14]	%I1.2
1-10A	15	ls_at[15]	%I1.3
1-10B	16	ls_at[16]	%I1.4
1-11	17	ls_at[17]	%I1.5
TRVL-4A (drop)	18	ls_at[18]	%I1.6 *(same as Loading)*
hoist up	—	ls_up	%I0.4
hoist in tank	—	ls_dip	%I0.5
start button	—	btn_start	%I2.1

Outputs (read): mv_up=%Q0.4 mv_down=%Q0.5 mv_rev(East)=%Q0.6 mv_fwd(West)=%Q0.7.

Direction: mv_fwd = **West** = toward higher station; mv_rev = **East** = toward lower station. Source of truth = ./IOMap.scl.